Interação Pessoa-Máquina 2026/2027

Lab Class 1 - User Interface Design Analysis

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1. Example of Good User Interface Design

Shazam Mobile App - Single-Button Home Screen ("Tap to Shazam")

1.1 Objective of the Whole Interface

Shazam is a mobile audio-recognition application whose overall objective is to identify music, TV shows, or advertisements playing in the user's immediate environment and provide the track title, artist name, lyrics, and streaming links.

1.2 Specific Good Aspects

Rather than greeting the user with a complex dashboard, navigation tabs, or search bars, the primary home screen of Shazam consists almost entirely of a single, large, central circular button labeled "Tap to Shazam" that subtly pulses on the screen. In addition, users can long-press this same central button to activate "Auto Shazam," which continuously listens for music in the background.

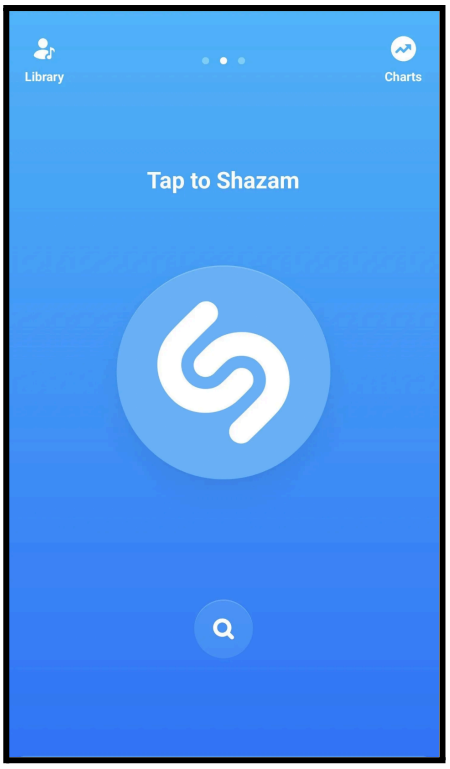
1.3 Justification (Why It Is Good)

- **Context-Aware Task Prioritization:** Music playing in a café, radio, or store is brief, it can end at any second. By making the identification button the immediate landing screen, the interface eliminates navigation friction and

allows the user to start recording within a fraction of a second of opening the app.

- **Fitts's Law Optimization:** According to Fitts's Law, the time required to move to a target area is a function of the distance to and size of the target. Occupying the center of the screen with an oversized button makes it effortless to tap with one thumb, even while walking, holding groceries, or barely looking at the phone.
- **Minimal Cognitive Load & Clear Affordance:** The pulsing logo mimics a physical button and visually communicates its interactive affordance ("press me"). By hiding secondary features (library, charts, concert listings) behind swipe gestures away from the main focal point, the interface prevents choice paralysis during a time-sensitive task.

1.4 Visual Illustration



The Shazam home screen centers the entire user experience around a single oversized, pulsing target, minimizing interaction time before a song ends.

2. Example of Bad User Interface Design

WhatsApp - "Delete for Me" vs. "Delete for Everyone" Confirmation Dialog

2.1 Objective of the Whole Interface

WhatsApp is a cross-platform instant messaging application that enables users to send text messages, voice notes, images, and documents to individuals and groups in real time.

2.2 Specific Bad Aspects

When a user selects a sent message and taps the trash can icon to delete it, WhatsApp presents a modal confirmation dialog containing three vertically stacked text buttons: "Delete for everyone," "Delete for me," and "Cancel." If a user who accidentally sent a sensitive or embarrassing message to the wrong group panics and mistakenly taps "Delete for me," the message is permanently removed from their own device's view while remaining completely visible to every other recipient in the chat which leaves the sender with no way to select the message again to delete it for everyone.

2.3 Justification (Why It Is Bad)

- **Destructive Trap & Loss of Control:** Tapping "Delete for me" on an outgoing message removes the user's local reference to that message without revoking it from the server. If the user misses the fleeting 5-second undo banner, the error becomes completely irreversible, trapping the message in the group chat forever.
- **Poor Visual Differentiation Under Stress:** Users attempting to unsend a message are typically acting under time pressure and anxiety (hoping to delete the message before others read it). Presenting two similarly styled text labels ("Delete for everyone" and "Delete for me") in close proximity inside the same compact modal violates Fitts's Law safety spacing and invites slip errors.
- **Ambiguous Mental Model:** When a user wants to retract a message they just sent by mistake, their primary mental model is "unsend." Mixing local housekeeping ("hide this from my screen") with global retraction ("unsend for all recipients") under the single label "Delete" blends two fundamentally different actions with opposite consequences.

2.4 Why It Was Likely Designed This Way

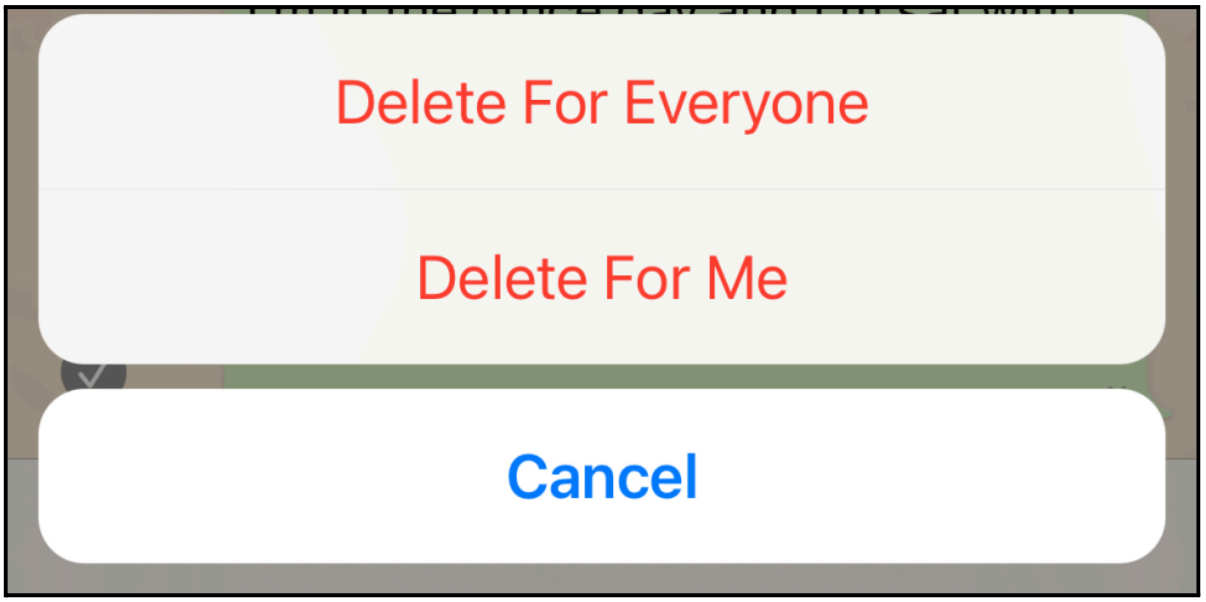
- **Feature Evolution & Legacy Architecture:** Historically, WhatsApp only supported local deletion ("Delete for me") to free up storage or tidy up a chat log. When global message retraction ("Delete for everyone") was introduced

later, developers simply appended the new option into the existing deletion dialog rather than redesigning the workflow into separate "Unsend" and "Clear Local Copy" actions.
- **UI Compactness:** Combining all deletion choices into a single standard system dialog box minimized engineering effort and kept the context menu short, even though it grouped a local view change with a critical network action.

2.5 Suggested Corrections and Improvements

- **Separate "Unsend" from "Delete Local Copy":** For recently sent messages within the revocation time window, replace the generic trash icon action with a primary **"Unsend Message"** action that defaults to removing it for everyone. Move "Remove from my device only" into a secondary submenu or require an explicit warning.
- **Allow Post-Deletion Recovery or Persistent Reference:** Instead of completely erasing the message bubble when "Delete for me" is clicked, leave a subtle placeholder visible only to the sender (e.g., *"You hid this message for yourself - Tap to Delete for Everyone"*), or keep the "Undo" option accessible for a longer period.
- **Visual Hierarchy & Secondary Confirmation:** Style "Delete for Everyone" as a prominent primary button and add a clear warning subtitle under "Delete for me" (*"Other people in the chat will still see this message"*) so users immediately recognize the consequence before tapping.

2.6 Visual Illustration



Stacking "Delete for me" directly next to "Delete for everyone" creates a high-stress slip hazard where hiding a message locally prevents the user from unsending it globally.